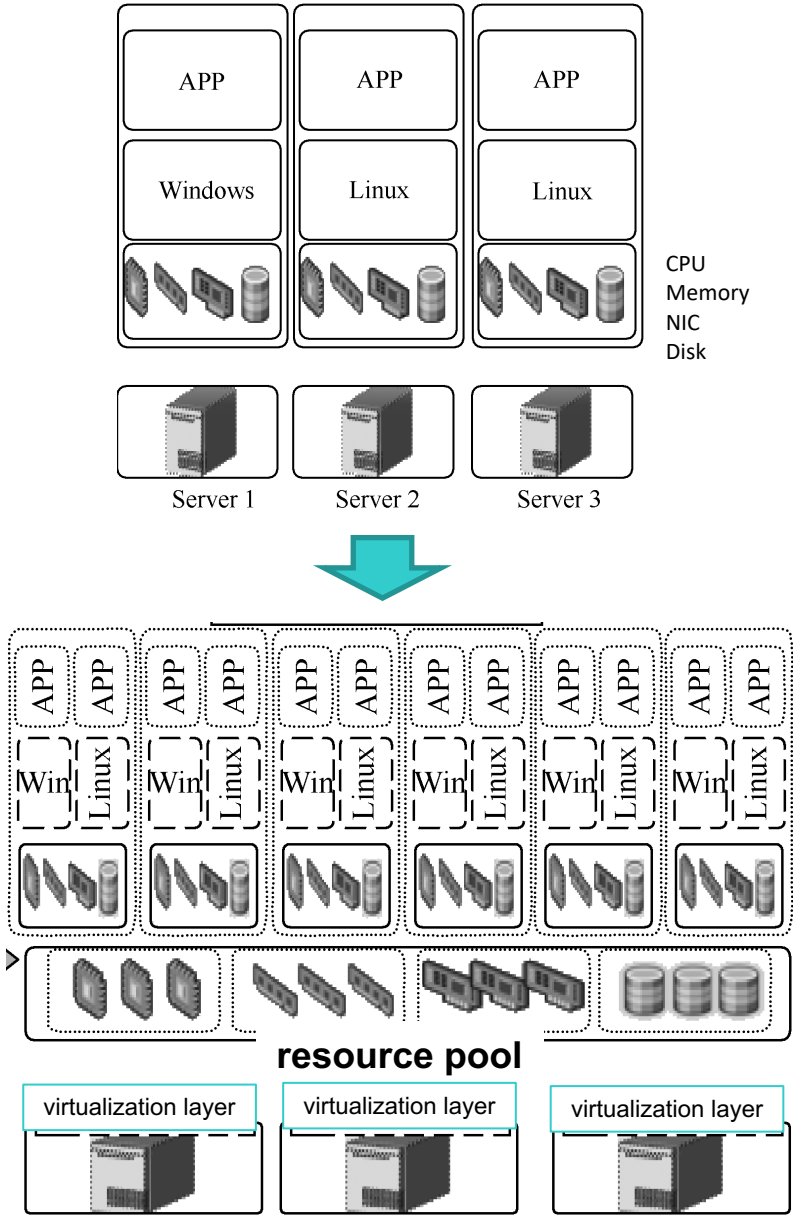
COMP4442

SERVICE AND CLOUD COMPUTING

Virtualization
Part-1

Virtualization

- Virtualization is the technology which allows multiple operating systems to run on a single physical system and share the underlying hardware resources
- The basic idea is to separate the hardware devices from the software framework
  - Eliminate the dependency of running software on the hardware environment
  - Maximize sharing and utilization of hardware resources
    - 15% --> 65+%



Virtualization is Abstraction

- Physical resources can be viewed as logical resources at a higher level of abstraction.
  - e.g., virtual memory in OS
  - Mask the complexity of physical devices and allow users to use and manage the shared resources in a centralized and unified way.
- Virtualization is applied at different levels, yielding
  - virtualized servers,
  - virtualized storage,
  - virtualized networks,
  - virtualized memory,
  - virtualized data,
  - virtualized applications,
  - virtualized desktops

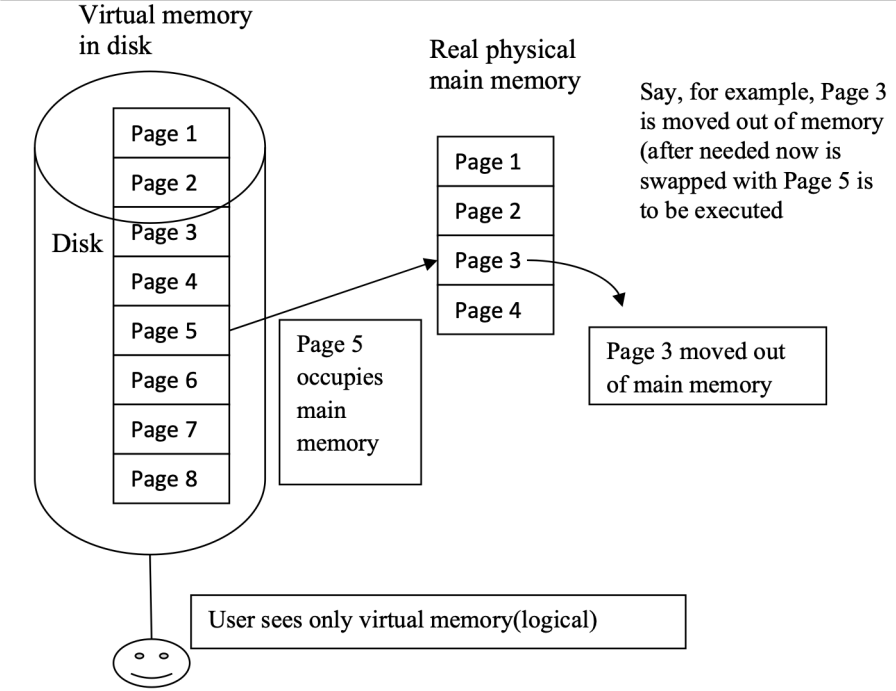
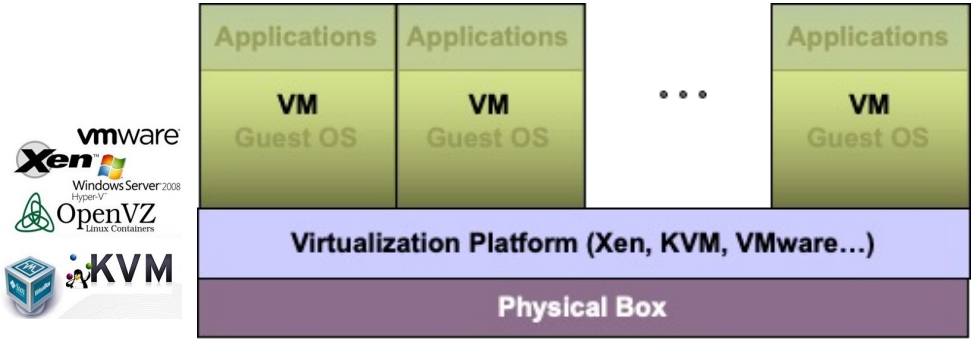


Fig. 3.5 Concept of virtual memory

Q&A: Write your own definition of virtualization

Virtualization is Abstraction : Hypervisors

- Made possible by the virtualization layer software called a *hypervisor* or *virtual machine monitor (VMM)*.
  - Emulate underlying hardware resources and presents a virtual operating platform to a guest OS.
  - Scheduler: fairly distributing processor resources among VMs, and targeting at maximum resource utilization
- Supporting multiple virtual machines is similar to running multiple processes simultaneously with a single CPU time-sharing them.
  - share the available hardware resources (processor, memory …) without conflict under hypervisor's control
  - Hypervisor *allocates* and *manages* the host machine's resources that are needed to each VM in turn and making sure that the VMS cannot disrupt each other



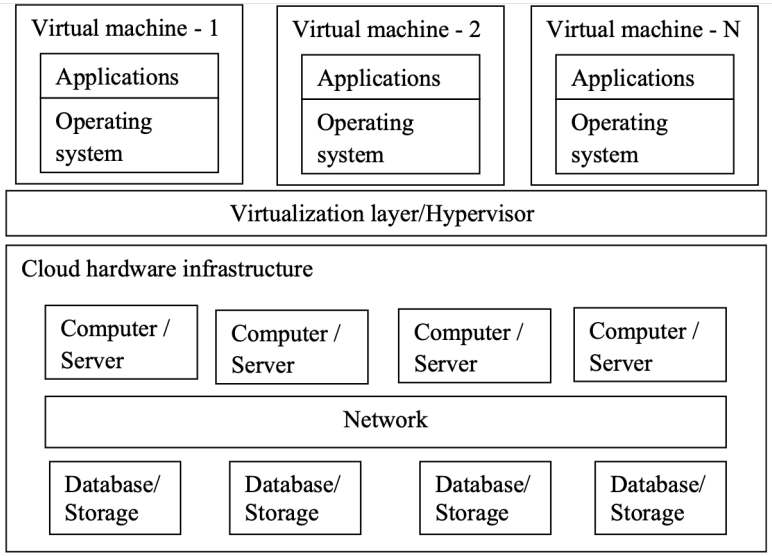


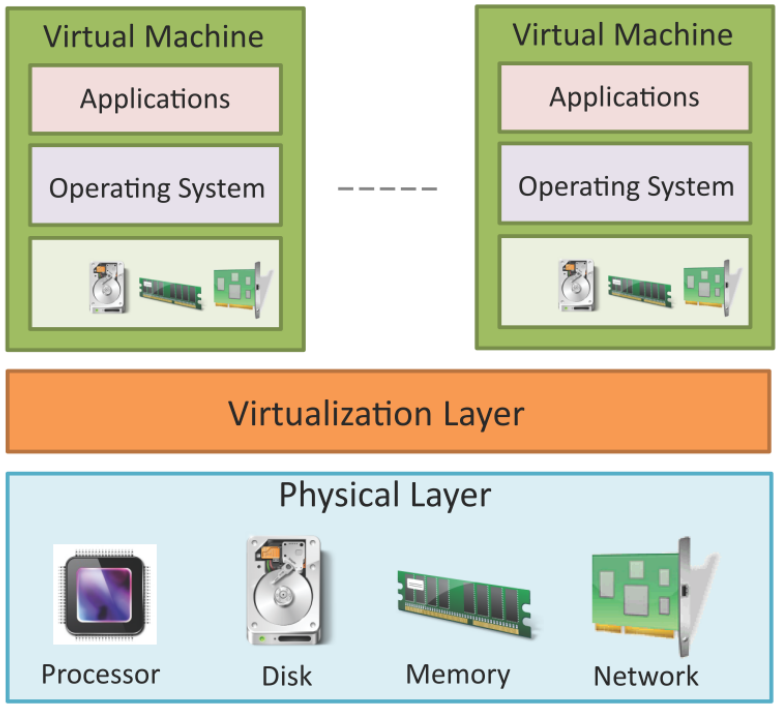
Fig. 3.8 Virtualized cloud environment

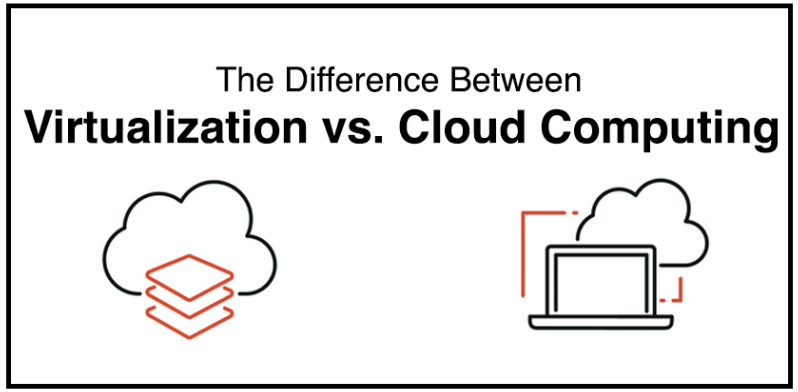
Properties of Virtualization

- Partitioning
  - Run multiple operating systems on one physical machine.
  - Divide system resources between virtual machines.
- Isolation
  - Provide fault and security isolation at the hardware level.
  - Preserve performance with advanced resource controls.
- Encapsulation
  - Save the entire state of a virtual machine to files.
  - Move and copy virtual machines as easily as moving and copying files.
- Hardware Independence
  - Provision or migrate any virtual machine to any physical server.

Virtualization vs. Cloud Computing

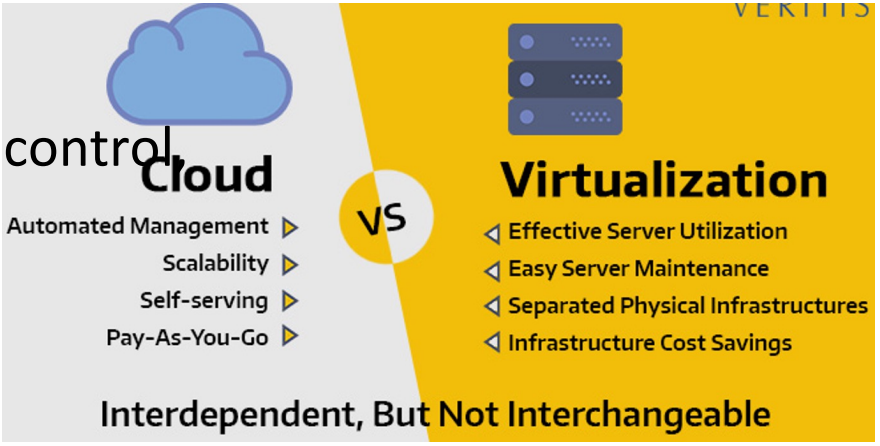
- Virtualization is the key enabling technology of cloud computing. It transfers the physical infrastructure into logical and virtual resources.
  - Enable sharing of resources by multiple users simultaneously, allowing them to have varying demands
  - Increase resource utilization and reduce cost.
- But what are the difference between virtualization technology and cloud computing?





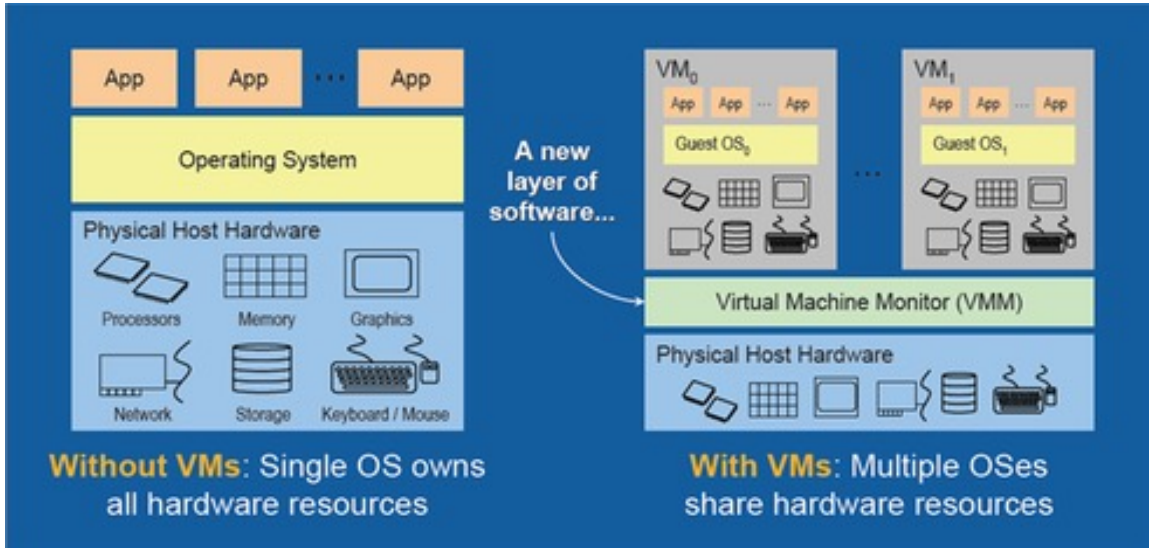
Virtualization vs. Cloud Computing

- Virtualization is just a software that provides us computing environments which is independent of physical infrastructure
- Cloud computing takes virtualization one step further:
  - You don’t need to own the hardware but get it as a service
  - Resources are rented on demand from a cloud
    - You get billed only for what you used
  - Various cloud service providers allow creating virtual servers:
    - Choose the OS and software each instance will have
    - The chosen OS will run on a large server farm
    - Instantiate or shut down virtual servers within minutes
  - Simple interfaces for management: submission, monitoring control,



Server Virtualization

- Many virtual servers on top of the same physical server infrastructure, logically separated from each other and unaware of the presence of each other.
  - Abstract the physical hardware into many virtual machines (VMs) to be provided to the users.
- A VM is a container holding a *guest OS* and applications.
  - A complete machine image that creates an isolated runtime environment
  - Loaded onto the hosting server and run like any other program.
  - Multiple VMs can run on a single physical machine



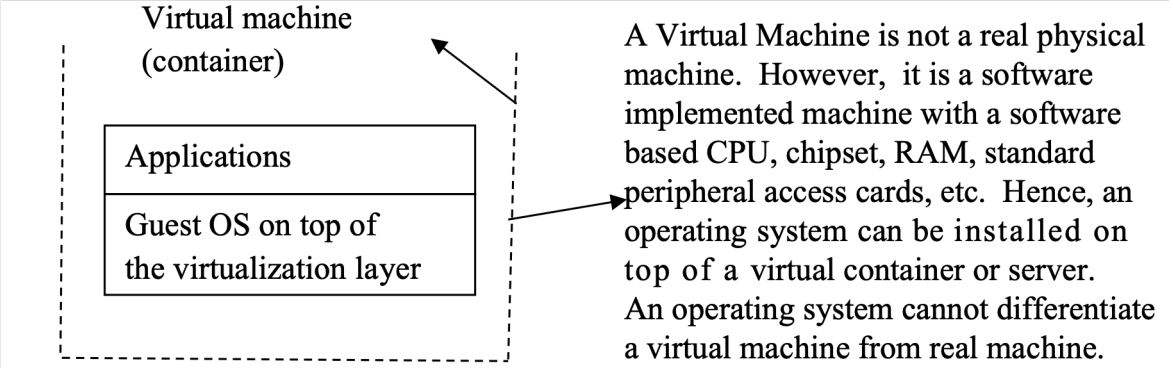


Fig. 3.9 Concept of virtual machine

Virtualization Software: Hypervisors

- Two types of Hypervisors
  - Type 1: Native (bare metal) hypervisors
    - run directly on the host hardware and control the hardware and monitor the guest OSs.
    - built as firmware within the hardware.
  - Type 2: Hosted hypervisors
    - run on top of a conventional (main/host) OS and monitor the guest OSs.
    - installed on top of the host OS.

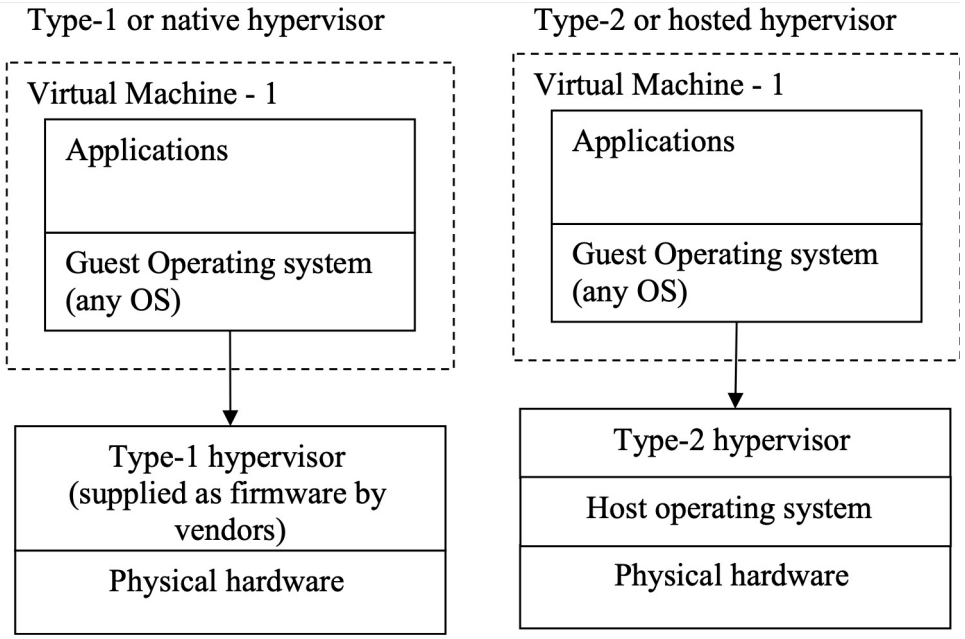


Fig. 3.10 Two types of hypervisors

- Comparisons
  - Type 1 has better performance than Type 2
  - Type 2 supports a wide range of hardware which can be produced by different vendors.
  - Installation of type 2 hardware is easier than type 1

Virtualization Methods

- 3 types of server virtualization methods with different applications requirements and costs.
  - Full virtualization
  - Para virtualization
  - OS level virtualization (containerization)

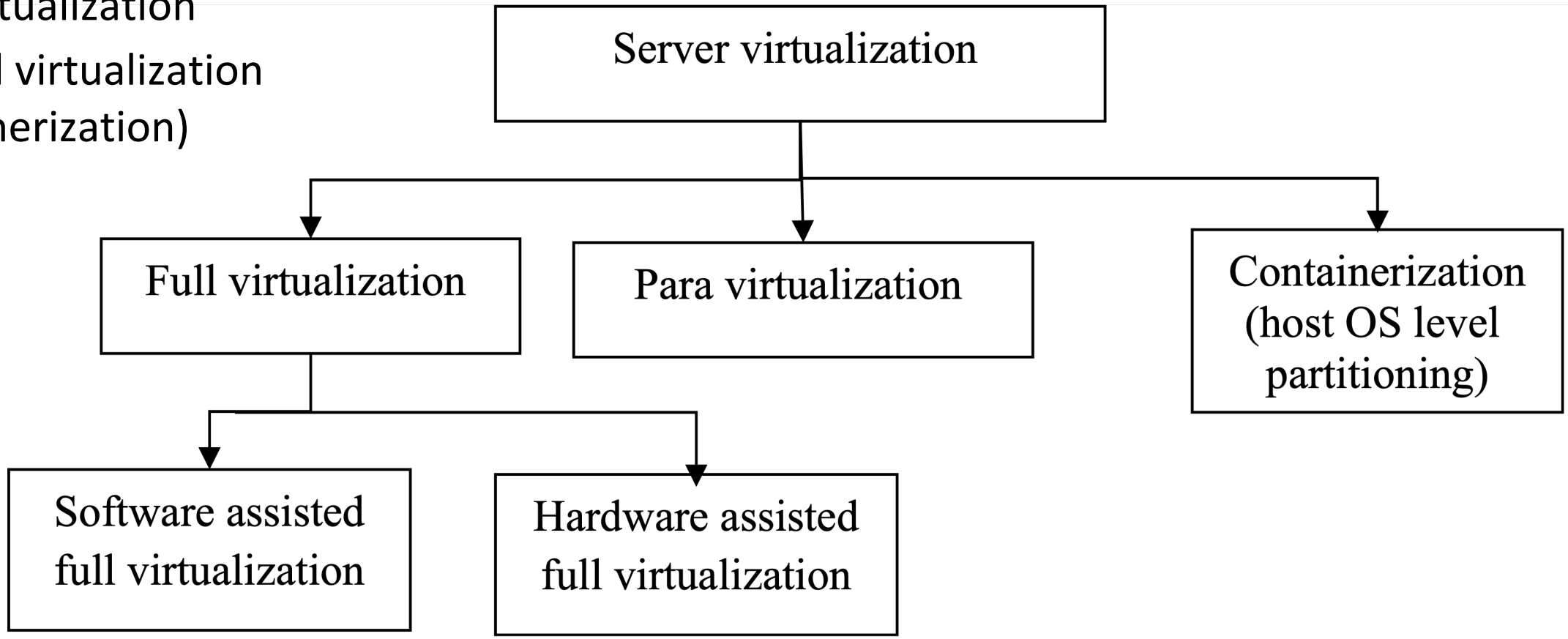


Fig. 3.11 Different types of server virtualization

Virtualization Methods: Full Virtualization

- Virtualization layer completely decouples the guest OS from the underlying hardware. Guest OS requires no modification for interacting with hardware
  - Guest OS runs as if it is the only OS running on top of the hardware and is unaware that it is being virtualized.
  - Enabled by combination of direct execution and binary translation of guest OS/app instructions.
- Software-assisted (ancient)
  - Guest OS virtual address -> Guest OS physical address –(translated by hypervisor)→ Host OS physical address.
  - Sensitive instructions (e.g. that needs to be run in the host OS's kernel mode) are dynamically translated by the hypervisor to safe trap routines.
  - Hardware interrupt forwarding: emulated by the hypervisor (needs VM-exit).
- Hardware-assisted (modern)
  - CPU is aware of virtualization, supports with special hardware mechanisms and instructions
  - Guest OS virtual address -> Guest OS physical address –(translated by CPU hardware: 2 stage page tables)→ Host OS physical address.
  - Sensitive instructions: conventional user mode (VMX ring 3), conventional kernel mode (VMX ring 0), and additional hypervisor mode (VMX root). Only VM critical sensitive instructions traps into the hypervisor mode.
  - Hardware interrupt forwarding: posted interrupts (no need of VM-exit).

Virtualization Methods: Para Virtualization

- Guest OS is aware of virtualization and modified to enable communication with hypervisor to improve performance and efficiency
  - Guest OS kernel is modified so that it can make direct hardware calls - replace non-virtualizable instructions with hyper-calls that communicate directly with the hypervisor
- Guest OS still has non-virtualizable part: virtual drivers for network and storage access for computation-intensive applications
  - virtual drivers' calls are privileged direct calls
  - other I/O calls (e.g., memory, interrupt, etc) are through hypervisor.
- Para virtualization is usually with open-source OSs like Linux; proprietary OSs like Windows will not permit modifications.

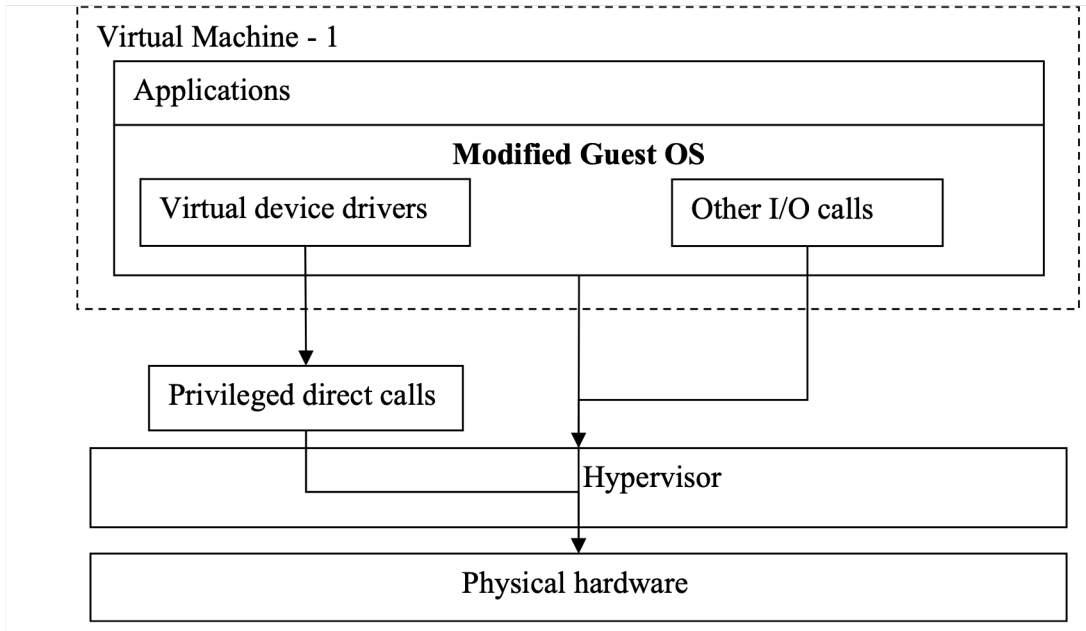


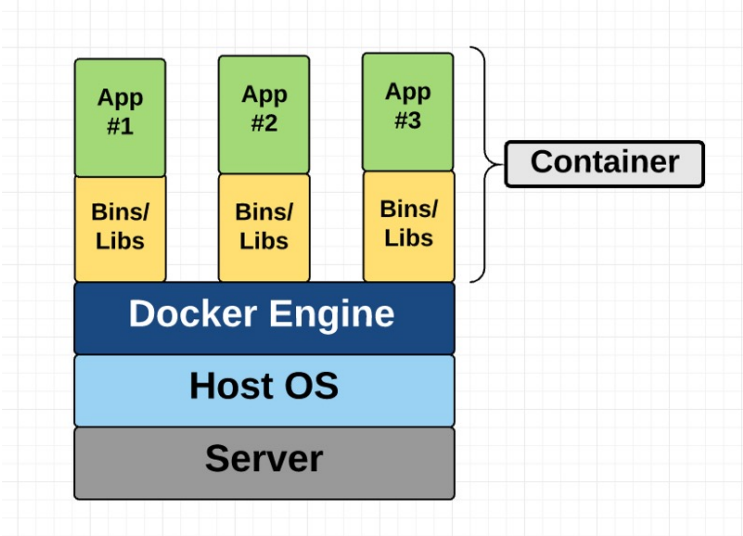
Fig. 3.14 Para virtualization

Virtualization Methods : Containerization

- Containerization is OS-level virtualization
  - Host OS creates many partitions or user spaces called containers.
    - Container packages up application and all of its library dependencies and data into a single, easy-to-manage unit
    - Rather than running a full OS, a container is layered on top of host OS and uses the OS's resources in a clever way.
  - There is no hypervisor. Host OS creates different containers for different users, and distributes its function to different applications
    - Eliminates overhead associated with hypervisors, giving a better performance than virtualization
- Docker is a container engine (runtime)
  - an open platform for developers and system admins to build, ship and run distributed applications

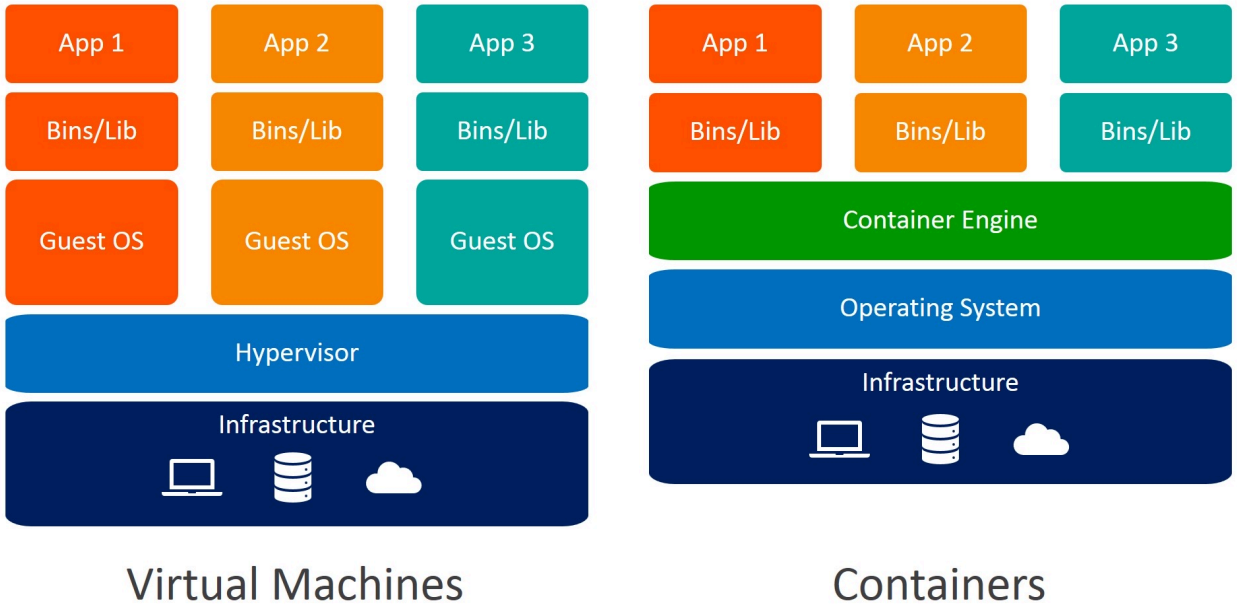
Application-1	Application-2	Application-3
Container	Container	Container
Host operating system		
Physical hardware/computer		

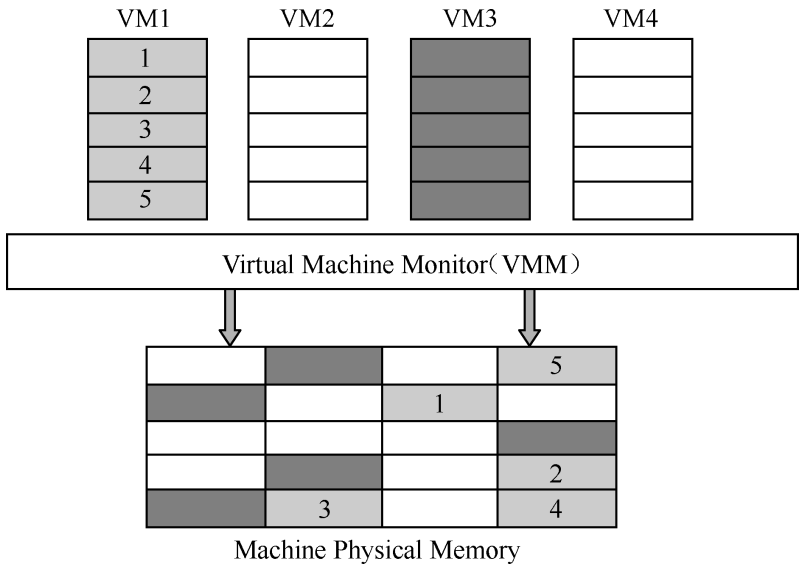
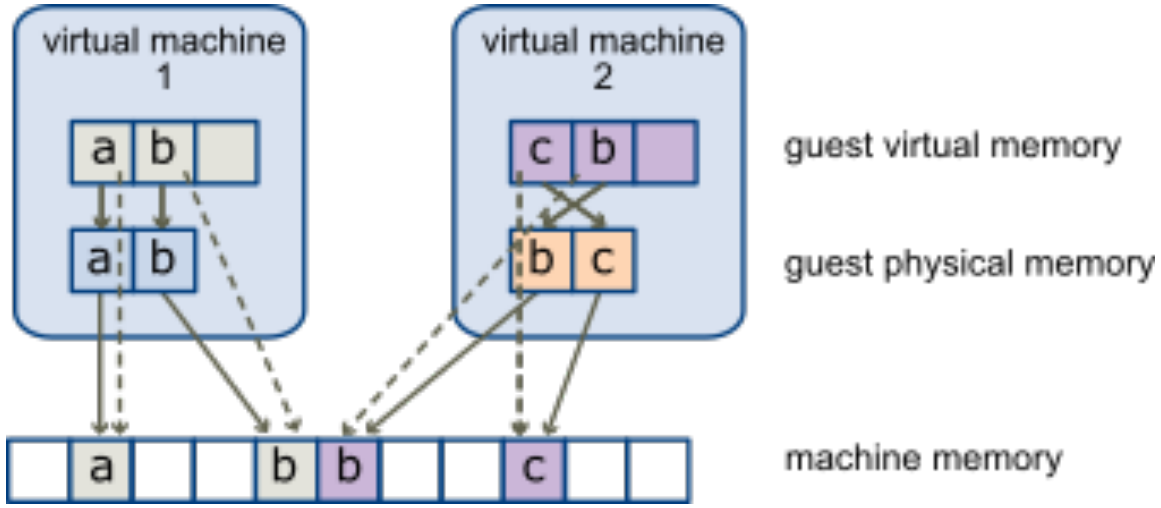
Fig. 3.15 Concept of containerization



Virtualization Methods : Containerization

- Containers vs VMs
  - Both are ways of deploying applications inside environments that are isolated from the underlying hardware. Key difference is the level of isolation.
  - Advantages of containers:
    - Lightweight and efficient (image typically mega-bytes; vs guest OS giga-bytes image)
    - Package and code reuse
    - Script-driven configuration – near-instant start time.
  - Disadvantages of containers
    - Process-level isolation, hence less secure



Memory Virtualization*

- Aggregate the physical primary memory of different servers into a single large primary memory
  - shared by applications deployed in VMs across the servers.
- 3 memory addresses:
  - guest virtual address that represents the virtual address as seen by the guest application.
  - guest physical address that represents the virtual address of memory as seen by the guest OS,
  - machine address that represents the actual physical address sent on the bus to memory